

Frame It as Cases: Portfolio Copy Deliverable

Voice Card

Curious, analytical, direct, conversational, precise, grounded.

Don't hedge unnecessarily. State the conclusion and the reasoning. Don't use filler or generic enthusiasm. Make caveats specific. Prefer concrete observations over vague claims. If a claim is strong, support it with the actual reasoning rather than softening it. Sound like a person thinking through a problem, not a person performing expertise.

Framed Sitemap

Note: Second result paragraph of selected work copy was added post peer-review of portfolio

1. Home

Purpose: Establish the claim quickly, give enough evidence to make it credible, and make the trial-project ask.

Content

- Short positioning/bio
- Agentic BI Builder
- Financial Risk Monitor
- Specific trial-project CTA

Home copy

I build end-to-end AI systems that connect data, models, and usable interfaces.

I'm strongest when I have to connect the pieces: data, models, APIs, workflows, and the interface that makes the result useful. My existing work includes AI, ML, and decision-support systems built through personal projects and capstones.

I'm now looking to apply that skill to real problems at early-stage companies. I want to work on problems that are still being figured out, build the system around what actually needs to happen, and iterate until it is actually useful.

Selected work

Agentic BI Builder

Starting a BI project means profiling the data, deciding what matters to the business, writing the queries, and building the report. That setup gets redone from scratch for every new business.

I built a working pipeline that turns raw business data into a Power BI project, with profiling, KPI inference, validation, and report generation handled end to end.

Financial Risk Monitor

Investors don't have one place to see portfolio risk alongside the macro conditions around it. That makes questions like “what happens if rates move?” slow to answer.

I built a portfolio risk tool that combines market and macro data, risk calculations, forecasts, and interactive scenario analysis in a 4-page Power BI dashboard.

CTA

Give me one problem you'd like solved by next week.

Give me one scoped problem. I'll spend a week building a solution and you can decide if you want me on the team.

Email me →

2. Projects

Purpose: Provide the complete evidence. Each project is framed as a short case with three beats: the problem, what I did and decided, and what came of it.

Order

1. Agentic BI Builder
2. Financial Risk Monitor
3. SolarSense
4. StudyGapAI
5. Alzheimer's Disease Classification

Agentic BI Builder

The problem. Starting a BI project means profiling the data, deciding what matters to the business, writing the queries, and building the report. That setup gets redone from scratch for every new business.

What I did. I built a BI report automation tool with deterministic data profiling and LLM-based KPI inference. Profiling is deterministic on purpose: typing and structuring a column has one correct answer and doesn't need an LLM guessing at it. KPI inference is where the judgment lives, so the LLM proposes KPIs within a schema rather than free text. A Reflector checks those proposals before they move forward. Early runs sometimes produced something technically valid but wrong for the data, like a churn rate for a business with no repeat customers. The final stage is a PBIR compiler I built by reverse-engineering Power BI's file format, so the output opens directly in Power BI Desktop rather than becoming another CSV to import manually.

What came of it. I ran the full pipeline against HR analytics, Chinook, and retail, three different domains, and all three completed end to end, with the pipeline catching and rectifying errors. Next time I'd test it against more domains before calling the architecture settled.

Financial Risk Monitor

The problem. Investors don't have one place to see portfolio risk alongside the macro conditions around it. That makes questions like “what happens if rates move?” slow to answer.

What I did. I built a portfolio risk monitoring tool that pulls live data from Yahoo Finance and FRED into SQLite when the update scripts run, computes VaR and drawdown, and runs 10,000-path Monte Carlo simulations. I chose Monte Carlo over historical VaR because historical VaR only shows what already happened. It can't answer what a 2 percentage point rate move would do to the portfolio. Prophet handles forecasting, and SQLite was enough because this is a single-analyst tool where concurrent writes weren't the actual constraint. I put the output into a 4-page Power BI dashboard with scenario slicers the investor can move themselves.

What came of it. Shipped as v1.0.0. The macro data currently informs the dashboard but doesn't feed directly into the risk calculations. Next time I'd make the macro data part of the risk calculations themselves.

SolarSense

The problem. A proper solar feasibility study takes 1 to 3 weeks of consultant time. For SMEs across Sub-Saharan Africa, that cost can put the decision out of reach before they even know whether solar makes sense.

What I did. As technical lead, I built the solar viability assessment engine behind the tool. It routes through a gateway to three services: one model for energy consumption, one for generation, and a deterministic ROI engine. ROI is arithmetic once the inputs exist, so I kept that part deterministic instead of adding a model where it adds variance without adding value. I trained both ML models myself, sourced CBECS, and reworked it for

African SME contexts since it wasn't built for them. I also built the shared authentication and rate limiting, and reviewed the security work on the parts I didn't personally build.

What came of it. Assessment time dropped from 1 to 3 weeks to about 3 minutes. Most cities came back with single-digit MAPE. Cape Town was roughly triple that because its Mediterranean climate didn't match the training distribution well. The models were trained on data from only 2 cities, which is the obvious limitation. I disclosed the Cape Town result rather than smoothing it over. Next time I'd expand the training data to more than 20 cities so the model holds up better on edge climates like Cape Town's.

StudyGapAI

The problem. Students can study hard and still fail because they're reinforcing what they already know while the actual gap causing the failure goes undiagnosed. A useful tool can't just grade the answer. It has to work out why the wrong answer happened.

What I did. I built a tool for diagnosing gaps and building a study plan around them, solo. It uses a React frontend, Flask backend, and Supabase for auth. It works in two steps: first it diagnoses the gap behind a wrong answer, then it builds a 6-week study plan around that gap. In one real case, it traced a wrong Calculus answer back to basic arithmetic and algebra prerequisites. The plan started there and moved toward Calculus once those dependencies were in place.

What came of it. It's live, deployed, and in use. I can explain how the plan is built, but I don't yet have data showing how much it improves results over time. Next time I'd track those outcomes systematically.

Alzheimer's Disease Classification

The problem. I built an Alzheimer's image classifier intended to assist with timely flagging, but a working classifier isn't enough. I needed to know which model to trust and whether its performance was actually real.

What I did. I trained and compared DenseNet, ResNet, and EfficientNet rather than defaulting to one architecture. For each model, I tested which blocks to freeze rather than tuning the same layers across all three. I also used per-image regularization. For model selection, I didn't optimize for the highest accuracy. I selected for the smallest training-validation gap because I wanted a better signal of generalization. Then I audited the dataset and found augmented duplicate scans across the train and test split, which meant the original result was leaked. I rejected it rather than reporting it.

What came of it. The three-architecture comparison and selection process are done. The corrected rerun on a clean dataset is still in progress, so I don't have a final performance number yet. Next time I'd audit the data thoroughly before building the elaborate pipeline, not after.

3. Resume

Purpose: Give visitors who already believe the claim a formal record of experience, education, and background.

No additional portfolio copy is required beyond a simple link to the resume.

View my resume →

4. Contact

Purpose: Give the target audience one clear, low-commitment way to act on the portfolio's proof.

CTA

Give me one problem you'd like solved by next week.

Give me one scoped problem. I'll spend a week building a solution and you can decide if you want me on the team.

Email me →

LinkedIn

→

GitHub →

Before / After

Generic AI line

I am a passionate and results-driven Applied AI engineer who leverages cutting-edge technologies to develop innovative, scalable solutions that solve complex real-world challenges.

Edited version

I care more about whether something actually solves the problem than how technically impressive the solution is.

The first describes the kind of person an AI engineer is expected to sound like. The second gives the reader an actual idea about how I approach the work.

Appendix – Old Copy (Pre-portfolio peer-review)

Home

I care more about whether something actually solves the problem than how technically impressive the solution is.

When I start a project, I first want to understand what the person actually needs, what the solution has to do, and what would make it useful. I haven't made a technical decision yet at that point. Once the problem is clear, the technical choices become much easier to reason about.

That's how I approach Applied AI. The technology matters, but only because it has to produce something useful in the real world. Sometimes that means a machine learning model. Sometimes it means an automated workflow, an AI system, or something much simpler.

I'm looking for an Applied AI role at an early-stage startup where I can work on problems that aren't already neatly mapped out. I want to build systems, own problems, and see the result actually get used.